

WHITFIELD FULFILLMENT

Warehouse Management System

Final Operations & Knowledge Handbook

Version 1.0 • Grounded Knowledge Source

Purpose

This handbook is the operational reference for the Whitfield Fulfillment WMS. It documents the supported warehouse workflows, permissions, inventory rules, receiving procedures, order and fulfillment flow, barcode behavior, audit requirements, and Voice AI safety rules.

Grounding rule

Knowledge Center answers must be based only on relevant material in this handbook. If the handbook does not contain enough information, the assistant must say so instead of inventing a procedure.

1. System Overview

Whitfield WMS is a centralized platform for managing warehouse inventory, products, sellers, receiving, customer orders, reservations, fulfillment, barcode scanning, users, warehouse access, and operational audit records.

The current warehouse locations are RENO and COLUMBUS.

Module	Purpose
Dashboard	Operational metrics and warehouse-level overview.
Inventory	Stock quantities, adjustments, reservations, and movements.
Products	Product, SKU, and UPC catalog information.
Sellers	Seller and merchant master data.
Receiving	Inbound shipment receiving and discrepancy recording.
Orders	Customer order creation, confirmation, and inventory reservation.
Fulfillment	Pick → Pack → Ship execution.
Barcode Scanner	UPC/barcode product identification.
Audit Trail	Operational history and change records.
Knowledge Center	Handbook-based operational guidance and citations.
Voice AI	Supported voice queries and confirmed operational commands.

2. Warehouse Scope & Access

WMS operations are warehouse-scoped. Non-administrator users may operate only within warehouses assigned to their accounts. The active warehouse selected in the application provides the operational context for supported screens and AI requests.

ADMIN: Global warehouse access and administrative permissions.

WAREHOUSE_MANAGER: Warehouse operations and management within assigned scope.

INVENTORY_CLERK: Supported inventory and warehouse operations within assigned scope.

READ_ONLY: Read-only access. Cannot perform inventory adjustments or other protected mutations.

Security rule: Backend authorization is authoritative. Frontend visibility is not a security boundary.

3. Inventory Management

Inventory records represent stock associated with a product and warehouse. Supported changes must preserve stock integrity and create movement history.

Inventory adjustment procedure

1. Open or locate the relevant inventory record.
2. Confirm the active warehouse and product/SKU.
3. Enter the required quantity adjustment.
4. Submit the adjustment.
5. Verify the resulting available quantity and movement history.

Negative-stock protection: An adjustment that would create an invalid negative available quantity must be rejected.

Reservations: Confirmed orders reserve available inventory. Orders requesting more stock than is available must not be confirmed.

Damaged stock: Damaged units must not be silently treated as normal available stock. They must be identified and recorded through the receiving/discrepancy process.

4. Receiving Operations

Receiving processes inbound shipments and records the quantity and condition of goods received at a warehouse.

Standard receiving procedure

1. Identify the inbound shipment and warehouse.
2. Inspect physical package condition.
3. Compare received quantities against expected quantities.
4. Record discrepancies instead of silently modifying inventory to hide the difference.
5. Separate damaged, crushed, or unsealed goods from normal stock.
6. Record damaged/discrepant quantities in the Receiving module.
7. Notify the shift/warehouse supervisor when escalation is required.

Damaged-item protocol: Inspect the item, place damaged/crushed/unsealed goods in the designated Yellow Quarantine Area, record the damaged/discrepant quantity in Receiving, and notify the shift supervisor. An RMA claim may be generated for the supplier when required.

Discrepancy rule: A quantity mismatch must be recorded as a discrepancy and must not be concealed by silently changing inventory.

5. Orders & Fulfillment

Customer orders are created against products and warehouse inventory. Confirmation requires sufficient inventory and protects against overselling.

Order workflow

Create order → Validate available inventory → Confirm order → Reserve inventory → Fulfill.

Fulfillment follows the controlled sequence READY_TO_PICK → PICKED → PACKED → SHIPPED.

Pick: Consumes reserved stock and records the PICK movement.

Pack: Advances the fulfillment task from PICKED to PACKED.

Ship: Assigns tracking information and changes the fulfillment/order status to SHIPPED.

State-machine protection: Invalid transitions, such as re-picking an already shipped task, must be rejected.

6. Barcode & UPC Scanning

The barcode scanner identifies products using barcode/UPC information and resolves the corresponding product/SKU.

1. Focus the scan input or use a physical keyboard-wedge scanner.
2. Scan the barcode/UPC.
3. Normalize the input and resolve the product.
4. Verify the returned SKU/product before an operational action.

UPC rule: UPC values are strings. Leading zeros are significant and must be preserved.

Unknown barcode: Return a not-found result and do not alter inventory.

7. Audit Trail

Protected operational actions are recorded in the audit trail. Inventory adjustments, order confirmation, fulfillment transitions, authentication events, and other supported system changes must remain traceable.

Inventory movement history and the audit trail serve complementary purposes: movement history describes stock movement, while the audit trail records operational system activity.

8. Voice AI & Voice Receiving

Voice AI supports read-oriented WMS questions and selected operational commands. Voice requests use the same backend authorization and warehouse-scope rules as normal application operations.

1. Read requests may return supported operational information.
2. Mutating commands require explicit confirmation before execution.
3. READ_ONLY users cannot bypass permissions through Voice AI.
4. Voice receiving produces structured receiving information and requires confirmation before the receiving mutation is submitted.
5. Voice AI does not directly access MongoDB; it uses authorized backend services/controllers.

9. Knowledge Center Grounding Rules

The Knowledge Center uses this handbook as its operational knowledge source. Answers should be concise, direct, and grounded in relevant handbook content.

Do not: dump unrelated handbook chunks, invent procedures, invent permissions, invent quantities, or use unrelated general knowledge as if it were WMS policy.

If information is insufficient: respond with: *The WMS handbook does not provide enough information to answer this question.*

Live-data separation: Questions about current inventory, current orders, or other changing operational data should use live WMS tools rather than handbook text.

Handbook separation: Questions about procedures, policies, permissions, and documented workflows should use this handbook.

10. Authoritative Operational Q&A;

Q: What is the receiving protocol for damaged items?

A: Inspect the physical package condition and compare received quantities against expected quantities. Separate damaged, crushed, or unsealed goods and place them in the designated Yellow Quarantine Area. Record the damaged/discrepant quantity in the Receiving module and notify the shift supervisor. An RMA claim may be generated for the supplier when required.

Q: What are the inventory quarantine and discrepancy rules?

A: Damaged or affected goods must be separated from normal stock according to the quarantine procedure. Receiving quantity mismatches must be recorded as discrepancies rather than silently changing inventory to hide the difference.

Q: How do I adjust inventory?

A: Open the relevant inventory record, confirm the warehouse and product, enter the quantity adjustment, and submit it. Verify the resulting available quantity and movement history. The adjustment must be authorized and must not create invalid negative available stock.

Q: Can a read-only user adjust inventory?

A: No. READ_ONLY users have read-only access and cannot perform inventory adjustments or other protected mutations.

Q: How does barcode scanning work?

A: The scanner accepts a barcode/UPC, normalizes the input, and resolves the matching product/SKU. UPC values are strings, so leading zeros must be preserved. Unknown barcodes must not change inventory.

Q: What is the order fulfillment process?

A: The fulfillment sequence is READY_TO_PICK → PICKED → PACKED → SHIPPED. Picking consumes reserved stock and records the PICK movement. Packing advances the task to PACKED. Shipping assigns tracking information and changes the fulfillment/order status to SHIPPED.

Q: Can an order be confirmed when inventory is insufficient?

A: No. The WMS protects against overselling. An order requesting more inventory than is available must be rejected.

Q: What happens when receiving quantities do not match expected quantities?

A: The discrepancy should be recorded in the Receiving module. The difference must not be concealed by silently modifying inventory.

Q: Can Voice AI directly modify the database?

A: No. Voice commands use the normal backend application and authorization layers. Mutating commands require explicit confirmation before execution.

Q: Can a read-only user bypass permissions using Voice AI?

A: No. Voice operations are subject to the same backend role and permission checks as normal application operations.

Q: How does warehouse access work?

A: Non-admin users can operate only within warehouses assigned to their accounts. The active warehouse provides operational context. ADMIN users have global warehouse access.

Q: What are the current warehouse locations?

A: The current WMS warehouse locations are RENO and COLUMBUS.

11. Insufficient Information & Out-of-Scope Questions

This handbook should not be used to answer unrelated general-knowledge questions or to invent operational procedures that are not documented here.

For questions outside the documented WMS knowledge, the safe response is: *The WMS handbook does not provide enough information to answer this question.*

12. Quick Reference

Topic	Rule
Damaged goods	Inspect → Yellow Quarantine Area → Record discrepancy → Notify supervisor.
Inventory adjustment	Authorized users only; update through Inventory; preserve movement/audit history.
READ_ONLY	Cannot perform protected mutations.
UPC	Treat as a string; preserve leading zeros.
Unknown barcode	Return not-found; do not modify stock.
Orders	Confirm only when sufficient inventory is available.
Fulfillment	READY_TO_PICK → PICKED → PACKED → SHIPPED.
Voice mutation	Authorization + explicit confirmation required.
Warehouse scope	Non-admin users operate only in assigned warehouses.
Audit	Supported protected operations remain traceable.

Document status: Final operational reference for the Whitfield WMS Knowledge Center. If application behavior changes, this handbook should be reviewed and re-indexed.